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Subject: Computer Vision Laboratory

Experiment - 6

Aim: To segment images considering both colour and spatial location.

Software Tools Required: VS Code, Python 3.12

Theory:

1. What is Image Segmentation?

Image segmentation is the process of dividing an image into meaningful regions (segments) that share similar properties such as color, texture, or intensity. It helps in object detection, recognition, and image analysis.

2. Why Colour Alone is Not Enough?

- Colour-based segmentation (e.g., using RGB/HSV values) works well for separating objects with distinct colours.
- But if two distant regions have the same colour, they may be wrongly grouped together.
- Example: In a picture of flowers, two red flowers at different positions might get segmented into one region if only colour is used.

3. Incorporating Spatial Location

To overcome this, spatial coordinates (x, y position of each pixel) are added along with colour information.

- Each pixel is represented as a feature vector:

$$F = [R, G, B, x, y]$$

or in other colour spaces:

$$\boldsymbol{F} = [\boldsymbol{L}, \boldsymbol{a}, \boldsymbol{b}, \boldsymbol{x}, \boldsymbol{y}]$$

(Lab space is often preferred because it is perceptually uniform.)

- This ensures that even if two regions have the same colour, they are treated as separate if they are spatially far apart.

4. Clustering-based Segmentation

Methods like K-Means clustering or Gaussian Mixture Models (GMMs) can be applied on this combined feature space:

- Each pixel's colour + location vector is clustered.
- Pixels in the same cluster are assigned to one segment.

5. Graph-based & Advanced Techniques

- Mean-Shift Clustering: Uses both colour and spatial proximity to form clusters.
- Normalized Cuts (Graph Cuts): Constructs a similarity graph based on colour and spatial closeness.
- Superpixel methods (SLIC): Partition images into compact regions using colour + spatial distance.

6. Practical Example

Suppose we want to segment an image of a red flower on a green background:

- Using only colour: all red pixels (flower petals) might be grouped together, even if scattered.
- Using colour + spatial info: red pixels close together (petal regions) will form one cluster, while distant red pixels (e.g., a red object elsewhere) form a different cluster.

7. Advantages

- More accurate segmentation of objects.
- Preserves spatial continuity (nearby pixels are grouped together).
- Helps in separating similar-coloured but spatially distinct objects.

Flowchart

Part-1: Image Segmentation using K-Means Clustering with Color and Spatial Features (BGR + XY)

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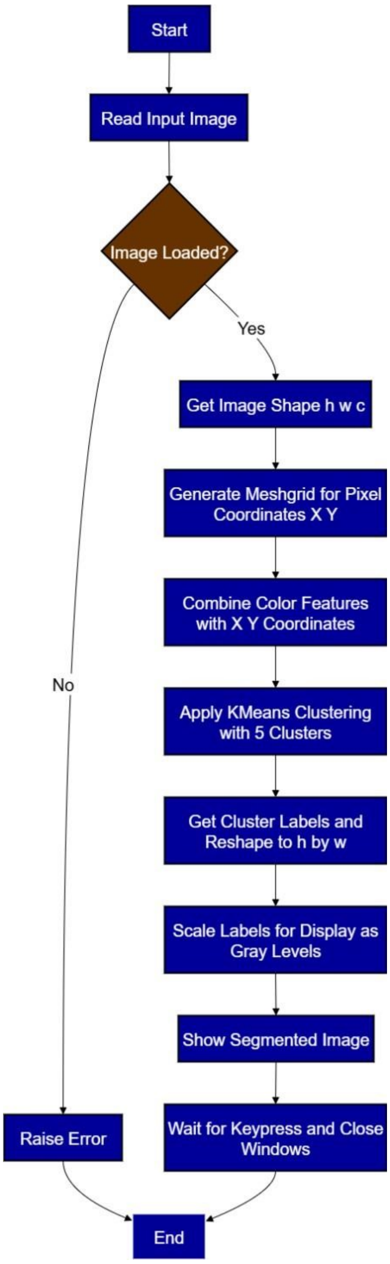

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Part-2: Color-Only vs. Color + Spatial Segmentation in K-Means

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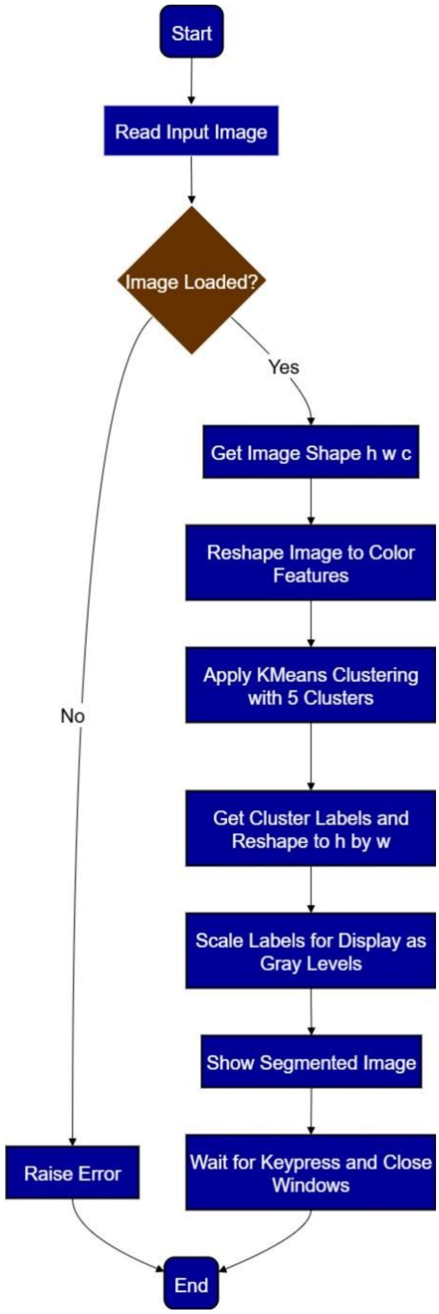

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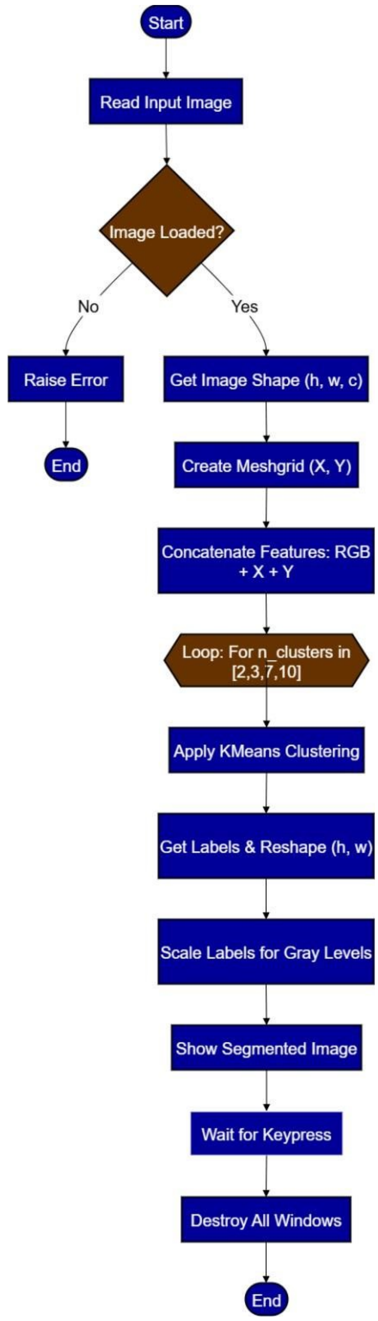
Part-3: Effect of Varying Number of Clusters on Image Segmentation



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Part 4: Interpretation of Cluster Centers in K-Means Segmentation

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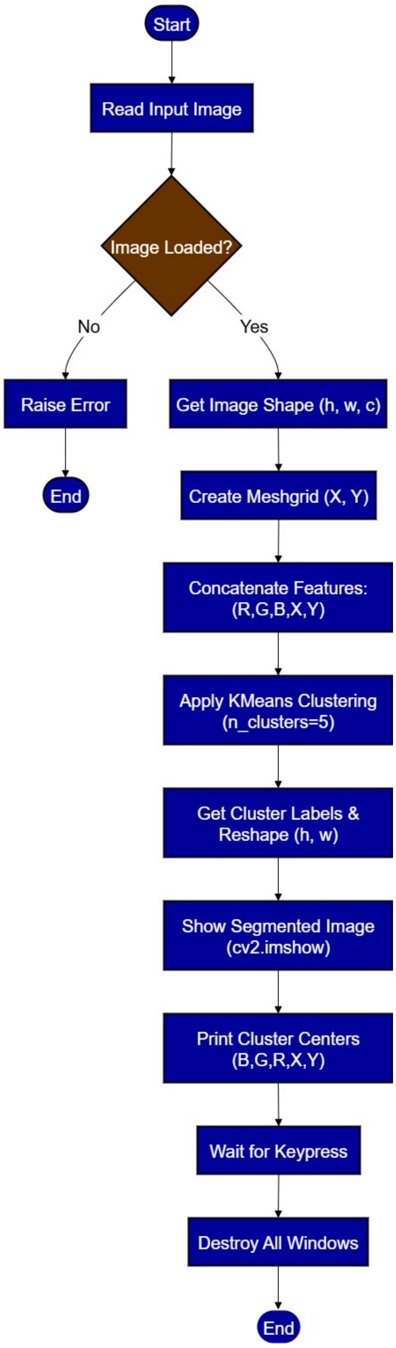

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Program with Output:

Program:

Part-1

```
import cv2
import numpy as np
from sklearn.cluster import KMeans

img = cv2.imread('building.jpg')
h, w, c = img.shape
X, Y = np.meshgrid(np.arange(w), np.arange(h))
features = np.concatenate([img.reshape((-1, 3)), X.reshape(-1, 1), Y.reshape(-1, 1)], axis=1)
kmeans = KMeans(n_clusters=5).fit(features)
labels = kmeans.labels_.reshape((h, w))
cv2.imshow('Segmented with XY', labels.astype('uint8') * 50)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Part-2

```
import cv2
import numpy as np
from sklearn.cluster import KMeans

img = cv2.imread('building.jpg')
h, w, c = img.shape

# Color-only clustering (no spatial coordinates)
features_color = img.reshape((-1, 3))
kmeans_color = KMeans(n_clusters=5, random_state=0).fit(features_color)
labels_color = kmeans_color.labels_.reshape((h, w))

cv2.imshow('Segmented (Color Only)', labels_color.astype('uint8') * 50)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

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Part-3

```python
import cv2
import numpy as np
from sklearn.cluster import KMeans

img = cv2.imread('building.jpg')
h, w, c = img.shape
X, Y = np.meshgrid(np.arange(w), np.arange(h))
features = np.concatenate([img.reshape((-1, 3)), X.reshape(-1, 1), Y.reshape(-1, 1)], axis=1)

for n_clusters in [2, 3, 7, 10]:
    kmeans = KMeans(n_clusters=n_clusters, random_state=0).fit(features)
    labels = kmeans.labels_.reshape((h, w))
    cv2.imshow(f'Segmented (Clusters={n_clusters})', labels.astype('uint8') * int(255/(n_clusters-1)))
    cv2.waitKey(0)

cv2.destroyAllWindows()
```

Part-4

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```
import cv2
import numpy as np
from sklearn.cluster import KMeans

img = cv2.imread('building.jpg')
h, w, c = img.shape
X, Y = np.meshgrid(np.arange(w), np.arange(h))
features = np.concatenate([img.reshape((-1, 3)), X.reshape(-1, 1), Y.reshape(-1, 1)], axis=1)
kmeans = KMeans(n_clusters=5).fit(features)
labels = kmeans.labels_.reshape((h, w))

# Visualize segmentation
cv2.imshow('Segmented with XY', labels.astype('uint8') * 50)

# Print cluster centers
print("Cluster centers (B, G, R, X, Y):")
for idx, center in enumerate(kmeans.cluster_centers_):
    b, g, r, x, y = center
    print(f"Cluster {idx+1}: B={b:.1f}, G={g:.1f}, R={r:.1f}, X={x:.1f}, Y={y:.1f}")

cv2.waitKey(0)
cv2.destroyAllWindows()
```


Output:

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Part-1

Part-2

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Part-3

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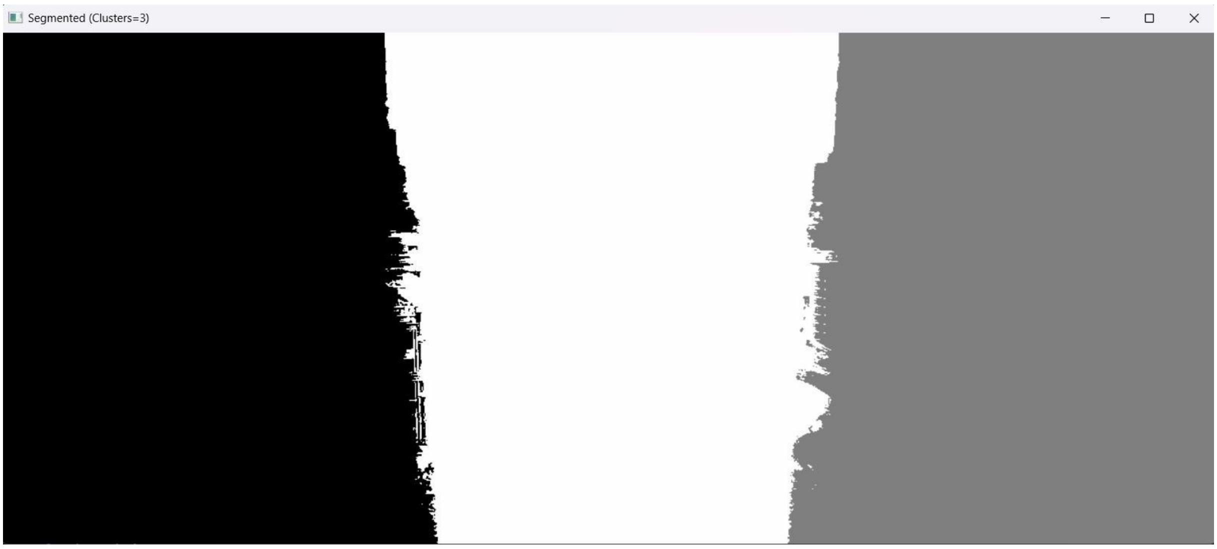

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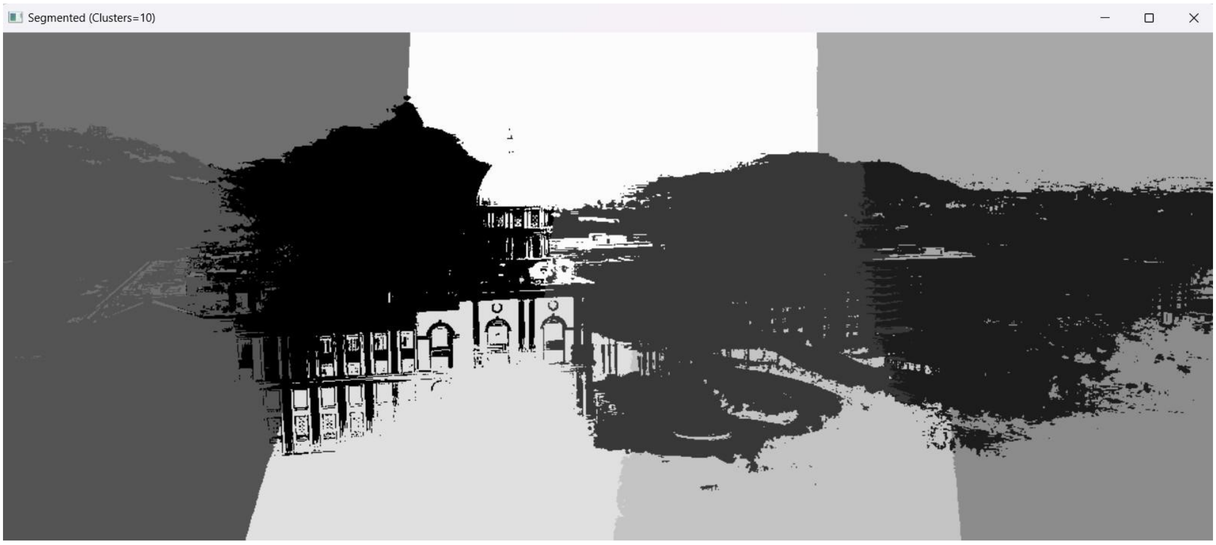

Part-4

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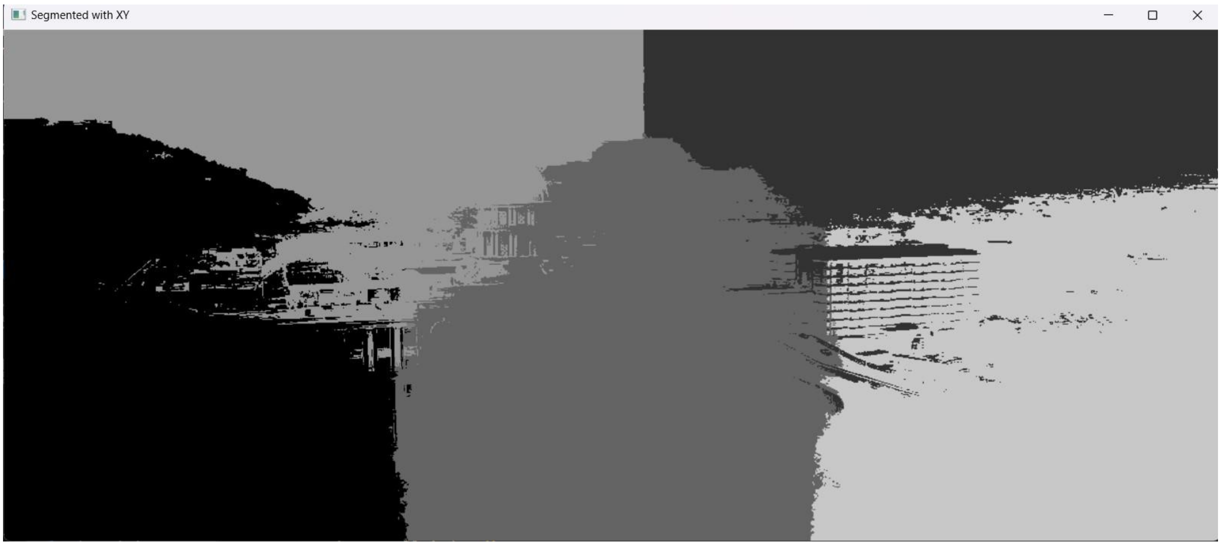
```
Cluster centers (B, G, R, X, Y):
Cluster 1: B=59.6, G=64.8, R=65.5, X=183.2, Y=344.2
Cluster 2: B=197.0, G=169.3, R=151.6, X=978.0, Y=103.4
Cluster 3: B=85.5, G=88.4, R=94.5, X=650.5, Y=353.3
Cluster 4: B=197.1, G=173.7, R=158.3, X=367.9, Y=110.4
Cluster 5: B=52.0, G=67.7, R=65.4, X=1083.9, Y=365.2
```

Conclusion/Inferences:

Image segmentation that incorporates both color and spatial location proves to be more effective and meaningful than using color information alone. By combining pixel intensity values with their spatial coordinates, segmentation algorithms such as K-Means are able to produce regions that are not only similar in appearance but also spatially coherent. This approach reduces the risk of scattered clusters, enhances object boundary detection, and leads to more accurate representation of real-world scenes. Thus, integrating color and spatial features creates a balanced segmentation technique suitable for practical applications in computer vision such as medical imaging, object recognition, and scene understanding.